

# Cost\_change\_source\_cluster\_documentation

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## R Markdown

Documentation for R script: Cost\_change\_source\_cluster.r which is called by "LASTING\_cluster\_w\_masterinput.r" and is used to calculate cost, environment and social pillar impact for a given dietary type and food source (food at home vs food away from home).

First, read in functions used to calculate impacts.

```
#Cost_change_source.r
#setwd("C:\\Users\\fcudhe01\\Box Sync\\new Life\\LASTING\\Code")
setwd("/cluster/tufts/lasting/shared/LASTING/Code")
source("cost_change_functions.r")
```

Define working directory and output location.

```
#file_location = "C:\\Users\\fcudhe01\\Box\\lasting_aim_3\\data\\in\\example_cra"
#file location already set in "LASTING.r"
setwd(file_location)

output_location = paste("/cluster/tufts/lasting/shared/model_development/data/outputs/cost_env/",
                        costs_output_loc.vec[k], sep="")
```

Read in relevant inputs (population draws, costs and impact factors of dietary factors) and do some minor data processing.

```
#pop.sims<-read.csv("population.sims.csv", fileEncoding = 'UTF-8-BOM')
pop.sims<-read.csv("observed.pop.draws.csv", fileEncoding = 'UTF-8-BOM')
#pop.sims$ subgroup_id<-pop.sims$ subgroup

cost.inputs<-costs_inputs.vec[[k]]
merged<-merge(cost.inputs, pop.sims, by="subgroup_id")
merged$Sex<-merged$Sex.y
merged$population <- merged$population.y
```

Create list objects to store outputs. Each element of the lists corresponds to a specific subgroup. There are 31 lists in total, corresponding so some combinaiton of 1. what we are looking at (impact (as in, impact of change), substitution impact, combined impact, population, delta (aka, change in intake), current intak eimpact, counterfactual intake impact), and 2. level of disaggregation (total, consumed, unconsumed, inedible, wasted). Hopefully, what each list corresponds to is clear from name.

```
#####
list.of.impact.sims.total<-list()
list.of.substitution.impact.sims.total<-list()
list.of.combined.impact.sims.total<-list()
list.of.population.sims<-list()
list.of.delta.sims.total<-list()

list.of.impact.sims.consumed<-list()
list.of.substitution.impact.sims.consumed<-list()
list.of.combined.impact.sims.consumed<-list()
list.of.delta.sims.consumed<-list()

list.of.impact.sims.unconsumed<-list()
list.of.substitution.impact.sims.unconsumed<-list()
list.of.combined.impact.sims.unconsumed<-list()
list.of.delta.sims.unconsumed<-list()

list.of.current.intake.impact.sims.total<-list()
list.of.CF.intake.impact.sims.total<-list()
list.of.current.intake.impact.sims.consumed<-list()
list.of.CF.intake.impact.sims.consumed<-list()
list.of.current.intake.impact.sims.unconsumed<-list()
list.of.CF.intake.impact.sims.unconsumed<-list()

list.of.impact.sims.inedible<-list()
list.of.substitution.impact.sims.inedible<-list()
list.of.combined.impact.sims.inedible<-list()
list.of.delta.sims.inedible<-list()

list.of.impact.sims.wasted<-list()
list.of.substitution.impact.sims.wasted<-list()
list.of.combined.impact.sims.wasted<-list()
list.of.delta.sims.wasted<-list()

list.of.current.intake.impact.sims.inedible<-list()
list.of.CF.intake.impact.sims.inedible<-list()
list.of.current.intake.impact.sims.wasted<-list()
list.of.CF.intake.impact.sims.wasted<-list()
```

Rename some variables.

```
n.sims<-nsim1
impact.factor.names<-econ.outcomes.vec
```

**Start for loop.** We are going to be traversing the “merged” file row by row and calculating impacts. Each row should correspond to a subgroup and dietary factor.

```
for(i in 1:dim(merged)[1]) ##for each row in the input file (corresponding to fcid for a given subgroup
{
```

Extract label variables and put them into their own data frames (to be merged later with the )

```
print(i)
vars.to.keep.for.df.a<-c("Foodgroup", "sex_gp", "race_gp", "age_gp",
                          "subgroup_id", "population",
                          "Mean_Intake", "SE_Intake", "CF_Mean_Intake", "CF_SE_Intake", "Intake_unit",

                          # "CED_unit", "GHG_unit", "WATER_unit", "BLUEWATER_unit", "Food_price_unit",
                          # "CED_to_intake_conversion",
                          # "GHG_to_intake_conversion",
                          # "WATER_to_intake_conversion",
                          # "BLUEWATER_to_intake_conversion",
                          # "Food_price_to_intake_conversion",
                          # "FL_to_intake_conversion",

                          paste0(all.outcomes.vec, "_unit"),
                          paste0(all.outcomes.vec, "_to_intake_conversion"))

df.a<-merged[i, vars.to.keep.for.df.a]
df_aa<-df.a[rep(seq_len(nrow(df.a)), each = length(impact.factor.names)),]
```

For the sake of readability of `code`, we extract various relevant inputs from `mergedp[i,]` and give them their own object with own name.

```

current.mean<-merged[i,"Mean_Intake"]
current.se<-merged[i,"SE_Intake"]
counterfactual.mean<-merged[i,"CF_Mean_Intake"]
counterfactual.se<-merged[i,"CF_SE_Intake"]
impact.factor.means<-merged[i,econ.outcomes.vec.mn]
impact.factor.ses<-merged[i,econ.outcomes.vec.se]
#RRunit<-merged[i, "price_to_intake_conversion"]

# idk why this isn't working
# RRunit<-merged[i, c("CED_to_intake_conversion", "GHG_to_intake_conversion",
#                     "WATER_to_intake_conversion", "BLUEWATER_to_intake_conversion",
#                     "Food_price_to_intake_conversion", "FL_to_intake_conversion")]

# new
RRunit <- merged %>% select(c(paste0(impact.factor.names, "_to_intake_conversion"))) %>%
  slice(1)

pop.sims.names<-paste("X", c(1:n.sims), sep="")
population.sims<-matrix(rep(x=as.numeric(merged[i, pop.sims.names])), times=length(impact.factor.names)
#population.sims<-as.numeric(merged[i, pop.sims.names])
#population<-merged[i, "population"]
#population.se<-merged[i, "population.se"]

substitution.impact.factor.means<-merged[i,econ.outcomes.substitution.vec.mn]
substitution.impact.factor.ses<-merged[i,econ.outcomes.substitution.vec.se]
substitution_unit<-merged[i,"substitution_unit"]

current_inedible_p<-merged[i,econ.current.inedible.p.mn]
current_inedible_p_se<-merged[i,econ.current.inedible.p.se]
current_foodwaste_p<-merged[i,econ.current.foodwaste.p.mn]
current_foodwaste_p_se<-merged[i,econ.current.foodwaste.p.se]
counterfactual_inedible_p<-merged[i,econ.counterfactual.inedible.p.mn]
counterfactual_inedible_p_se<-merged[i,econ.counterfactual.inedible.p.se]
counterfactual_foodwaste_p<-merged[i,econ.counterfactual.foodwaste.p.mn]
counterfactual_foodwaste_p_se<-merged[i,econ.counterfactual.foodwaste.p.se]

```

take the above inputs and feed it into `function` that simulates the impact. The resulting “b” is a list of 31 matrices with simulation results that correspond to the 31 lists we made earlier.

```
b<-simulate.impact(current.mean=current.mean, current.se=current.se, counterfactual.mean=counterfactu
  counterfactual.se=counterfactual.se, impact.factor.names=impact.factor.names,
  impact.factor.means=as.numeric(impact.factor.means), impact.factor.ses=as.numeric(
  RRunit=RRunit,
  substitution.impact.factor.means=as.numeric(substitution.impact.factor.means),
  substitution.impact.factor.ses=as.numeric(substitution.impact.factor.ses),
  substitution_unit=substitution_unit,
  current_inedible_p=as.numeric(current_inedible_p),
  current_inedible_p_se=as.numeric(current_inedible_p_se),
  current_foodwaste_p=as.numeric(current_foodwaste_p),
  current_foodwaste_p_se=as.numeric(current_foodwaste_p_se),
  counterfactual_inedible_p=as.numeric(counterfactual_inedible_p),
  counterfactual_inedible_p_se=as.numeric(counterfactual_inedible_p_se),
  counterfactual_foodwaste_p=as.numeric(counterfactual_foodwaste_p),
  counterfactual_foodwaste_p_se=as.numeric(counterfactual_foodwaste_p_se),
  n.sims=n.sims,
  population.sims=population.sims)
```

Extract results from “b” (which itself is a list of 31 matrices) and create data frames with proper labels.

```
df.b1<-data.frame(outcome=rownames(b[[1]]), b[[1]])
df.b2<-data.frame(outcome=rownames(b[[2]]), b[[2]])
df.b3<-data.frame(outcome=rownames(b[[3]]), b[[3]])
df.b4<-data.frame(outcome=rownames(b[[4]]), b[[4]])
df.b5<-data.frame(outcome=rownames(b[[5]]), b[[5]])

df.b6<-data.frame(outcome=rownames(b[[6]]), b[[6]])
df.b7<-data.frame(outcome=rownames(b[[7]]), b[[7]])
df.b8<-data.frame(outcome=rownames(b[[8]]), b[[8]])
df.b9<-data.frame(outcome=rownames(b[[9]]), b[[9]])

df.b10<-data.frame(outcome=rownames(b[[10]]), b[[10]])
df.b11<-data.frame(outcome=rownames(b[[11]]), b[[11]])
df.b12<-data.frame(outcome=rownames(b[[12]]), b[[12]])
df.b13<-data.frame(outcome=rownames(b[[13]]), b[[13]])

df.b14<-data.frame(outcome=rownames(b[[14]]), b[[14]])
df.b15<-data.frame(outcome=rownames(b[[15]]), b[[15]])
df.b16<-data.frame(outcome=rownames(b[[16]]), b[[16]])
df.b17<-data.frame(outcome=rownames(b[[17]]), b[[17]])
df.b18<-data.frame(outcome=rownames(b[[18]]), b[[18]])
df.b19<-data.frame(outcome=rownames(b[[19]]), b[[19]])

df.b20<-data.frame(outcome=rownames(b[[20]]), b[[20]])
df.b21<-data.frame(outcome=rownames(b[[21]]), b[[21]])
df.b22<-data.frame(outcome=rownames(b[[22]]), b[[22]])
df.b23<-data.frame(outcome=rownames(b[[23]]), b[[23]])

df.b24<-data.frame(outcome=rownames(b[[24]]), b[[24]])
df.b25<-data.frame(outcome=rownames(b[[25]]), b[[25]])
df.b26<-data.frame(outcome=rownames(b[[26]]), b[[26]])
df.b27<-data.frame(outcome=rownames(b[[27]]), b[[27]])

df.b28<-data.frame(outcome=rownames(b[[28]]), b[[28]])
df.b29<-data.frame(outcome=rownames(b[[29]]), b[[29]])
df.b30<-data.frame(outcome=rownames(b[[30]]), b[[30]])
df.b31<-data.frame(outcome=rownames(b[[31]]), b[[31]])

df.c1<-cbind(df.aa, df.b1)
df.c2<-cbind(df.aa, df.b2)
df.c3<-cbind(df.aa, df.b3)
df.c4<-cbind(df.aa, df.b4)
df.c5<-cbind(df.aa, df.b5)

df.c6<-cbind(df.aa, df.b6)
df.c7<-cbind(df.aa, df.b7)
df.c8<-cbind(df.aa, df.b8)
df.c9<-cbind(df.aa, df.b9)

df.c10<-cbind(df.aa, df.b10)
df.c11<-cbind(df.aa, df.b11)
df.c12<-cbind(df.aa, df.b12)
df.c13<-cbind(df.aa, df.b13)
```

```
df.c14<-cbind(df.aa, df.b14)
df.c15<-cbind(df.aa, df.b15)
df.c16<-cbind(df.aa, df.b16)
df.c17<-cbind(df.aa, df.b17)
df.c18<-cbind(df.aa, df.b18)
df.c19<-cbind(df.aa, df.b19)

df.c20<-cbind(df.aa, df.b20)
df.c21<-cbind(df.aa, df.b21)
df.c22<-cbind(df.aa, df.b22)
df.c23<-cbind(df.aa, df.b23)

df.c24<-cbind(df.aa, df.b24)
df.c25<-cbind(df.aa, df.b25)
df.c26<-cbind(df.aa, df.b26)
df.c27<-cbind(df.aa, df.b27)

df.c28<-cbind(df.aa, df.b28)
df.c29<-cbind(df.aa, df.b29)
df.c30<-cbind(df.aa, df.b30)
df.c31<-cbind(df.aa, df.b31)

impact_unit_strings<-paste(df.b1$outcome, "_unit", sep="")

df.c1$outcome_unit<-NA
df.c2$outcome_unit<-NA
df.c3$outcome_unit<-NA
df.c4$outcome_unit<-NA
df.c5$outcome_unit<-NA

df.c6$outcome_unit<-NA
df.c7$outcome_unit<-NA
df.c8$outcome_unit<-NA
df.c9$outcome_unit<-NA

df.c10$outcome_unit<-NA
df.c11$outcome_unit<-NA
df.c12$outcome_unit<-NA
df.c13$outcome_unit<-NA

df.c14$outcome_unit<-NA
df.c15$outcome_unit<-NA
df.c16$outcome_unit<-NA
df.c17$outcome_unit<-NA
df.c18$outcome_unit<-NA
df.c19$outcome_unit<-NA

df.c20$outcome_unit<-NA
df.c21$outcome_unit<-NA
df.c22$outcome_unit<-NA
df.c23$outcome_unit<-NA

df.c24$outcome_unit<-NA
df.c25$outcome_unit<-NA
```

```
df.c26$outcome_unit<-NA
df.c27$outcome_unit<-NA

df.c28$outcome_unit<-NA
df.c29$outcome_unit<-NA
df.c30$outcome_unit<-NA
df.c31$outcome_unit<-NA
```

```
for(j in 1:dim(df.c1)[1])
{
  df.c1$outcome_unit[j]<-df.c1[j,impact_unit_strings[j]]
  df.c2$outcome_unit[j]<-df.c2[j,impact_unit_strings[j]]
  df.c3$outcome_unit[j]<-df.c3[j,impact_unit_strings[j]]
  df.c4$outcome_unit[j]<-df.c4[j,impact_unit_strings[j]]
  df.c5$outcome_unit[j]<-df.c5[j,impact_unit_strings[j]]

  df.c6$outcome_unit[j]<-df.c6[j,impact_unit_strings[j]]
  df.c7$outcome_unit[j]<-df.c7[j,impact_unit_strings[j]]
  df.c8$outcome_unit[j]<-df.c8[j,impact_unit_strings[j]]
  df.c9$outcome_unit[j]<-df.c9[j,impact_unit_strings[j]]

  df.c10$outcome_unit[j]<-df.c10[j,impact_unit_strings[j]]
  df.c11$outcome_unit[j]<-df.c11[j,impact_unit_strings[j]]
  df.c12$outcome_unit[j]<-df.c12[j,impact_unit_strings[j]]
  df.c13$outcome_unit[j]<-df.c13[j,impact_unit_strings[j]]

  df.c14$outcome_unit[j]<-df.c14[j,impact_unit_strings[j]]
  df.c15$outcome_unit[j]<-df.c15[j,impact_unit_strings[j]]
  df.c16$outcome_unit[j]<-df.c16[j,impact_unit_strings[j]]
  df.c17$outcome_unit[j]<-df.c17[j,impact_unit_strings[j]]
  df.c18$outcome_unit[j]<-df.c18[j,impact_unit_strings[j]]
  df.c19$outcome_unit[j]<-df.c19[j,impact_unit_strings[j]]

  df.c20$outcome_unit[j]<-df.c20[j,impact_unit_strings[j]]
  df.c21$outcome_unit[j]<-df.c21[j,impact_unit_strings[j]]
  df.c22$outcome_unit[j]<-df.c22[j,impact_unit_strings[j]]
  df.c23$outcome_unit[j]<-df.c23[j,impact_unit_strings[j]]

  df.c24$outcome_unit[j]<-df.c24[j,impact_unit_strings[j]]
  df.c25$outcome_unit[j]<-df.c25[j,impact_unit_strings[j]]
  df.c26$outcome_unit[j]<-df.c26[j,impact_unit_strings[j]]
  df.c27$outcome_unit[j]<-df.c27[j,impact_unit_strings[j]]

  df.c28$outcome_unit[j]<-df.c28[j,impact_unit_strings[j]]
  df.c29$outcome_unit[j]<-df.c29[j,impact_unit_strings[j]]
  df.c30$outcome_unit[j]<-df.c30[j,impact_unit_strings[j]]
  df.c31$outcome_unit[j]<-df.c31[j,impact_unit_strings[j]]
}
df.c1<- df.c1 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings) ##move the
df.c2<- df.c2 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c3<- df.c3 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c4<- df.c4 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c5<- df.c5 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c6<- df.c6 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
```



```
df.c7<- df.c7 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c8<- df.c8 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c9<- df.c9 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c10<- df.c10 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c11<- df.c11 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c12<- df.c12 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c13<- df.c13 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c14<- df.c14 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c15<- df.c15 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c16<- df.c16 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c17<- df.c17 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c18<- df.c18 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c19<- df.c19 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c20<- df.c20 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c21<- df.c21 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c22<- df.c22 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c23<- df.c23 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c24<- df.c24 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c25<- df.c25 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c26<- df.c26 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c27<- df.c27 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c28<- df.c28 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c29<- df.c29 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c30<- df.c30 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c31<- df.c31 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
```

We assign each dataframe we made to be the i-th element of their corresponding lists.

```

list.of.impact.sims.total[[i]]<-df.c1
list.of.substitution.impact.sims.total[[i]]<-df.c2
list.of.combined.impact.sims.total[[i]]<-df.c3
list.of.population.sims[[i]]<-df.c4
list.of.delta.sims.total[[i]]<-df.c5

list.of.impact.sims.consumed[[i]]<-df.c6
list.of.substitution.impact.sims.consumed[[i]]<-df.c7
list.of.combined.impact.sims.consumed[[i]]<-df.c8
list.of.delta.sims.consumed[[i]]<-df.c9

list.of.impact.sims.unconsumed[[i]]<-df.c10
list.of.substitution.impact.sims.unconsumed[[i]]<-df.c11
list.of.combined.impact.sims.unconsumed[[i]]<-df.c12
list.of.delta.sims.unconsumed[[i]]<-df.c13

list.of.current.intake.impact.sims.total[[i]]<-df.c14
list.of.CF.intake.impact.sims.total[[i]]<-df.c15
list.of.current.intake.impact.sims.consumed[[i]]<-df.c16
list.of.CF.intake.impact.sims.consumed[[i]]<-df.c17
list.of.current.intake.impact.sims.unconsumed[[i]]<-df.c18
list.of.CF.intake.impact.sims.unconsumed[[i]]<-df.c19

list.of.impact.sims.inedible[[i]]<-df.c20
list.of.substitution.impact.sims.inedible[[i]]<-df.c21
list.of.combined.impact.sims.inedible[[i]]<-df.c22
list.of.delta.sims.inedible[[i]]<-df.c23

list.of.impact.sims.wasted[[i]]<-df.c24
list.of.substitution.impact.sims.wasted[[i]]<-df.c25
list.of.combined.impact.sims.wasted[[i]]<-df.c26
list.of.delta.sims.wasted[[i]]<-df.c27

list.of.current.intake.impact.sims.inedible[[i]]<-df.c28
list.of.CF.intake.impact.sims.inedible[[i]]<-df.c29
list.of.current.intake.impact.sims.wasted[[i]]<-df.c30
list.of.CF.intake.impact.sims.wasted[[i]]<-df.c31

```

End the loop.

```

}
```

For each list, row-bind the elements of the **list** (corresponding to a subgroup and dietary factor) together into a data frame. Each of these dataframes has results for all dietary factors and subgroups, for one outcome (specified in name of dataframe). NAs are also replaced with 0.

```

##not sure why I went with full joint before, rbind is much faster, and shouldn't cause problems.

# added NaN removal on 10/6
impact.sims.total.allgroups<-Reduce(rbind,list.of.impact.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
substitution.impact.sims.total.allgroups<-Reduce(rbind,list.of.substitution.impact.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
combined.impact.sims.total.allgroups<-Reduce(rbind,list.of.combined.impact.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
population.sims.allgroups<-Reduce(rbind,list.of.population.sims) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
delta.sims.total.allgroups<-Reduce(rbind,list.of.delta.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

impact.sims.consumed.allgroups<-Reduce(rbind,list.of.impact.sims.consumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
substitution.impact.sims.consumed.allgroups<-Reduce(rbind,list.of.substitution.impact.sims.consumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
combined.impact.sims.consumed.allgroups<-Reduce(rbind,list.of.combined.impact.sims.consumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
delta.sims.consumed.allgroups<-Reduce(rbind,list.of.delta.sims.consumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

impact.sims.unconsumed.allgroups<-Reduce(rbind,list.of.impact.sims.unconsumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
substitution.impact.sims.unconsumed.allgroups<-Reduce(rbind,list.of.substitution.impact.sims.unconsumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
combined.impact.sims.unconsumed.allgroups<-Reduce(rbind,list.of.combined.impact.sims.unconsumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
delta.sims.unconsumed.allgroups<-Reduce(rbind,list.of.delta.sims.unconsumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

current.intake.impact.sims.total.allgroups<-Reduce(rbind,list.of.current.intake.impact.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
CF.intake.impact.sims.total.allgroups<-Reduce(rbind,list.of.CF.intake.impact.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
current.intake.impact.sims.consumed.allgroups<-Reduce(rbind,list.of.current.intake.impact.sims.consumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
CF.intake.impact.sims.consumed.allgroups<-Reduce(rbind,list.of.CF.intake.impact.sims.consumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
current.intake.impact.sims.unconsumed.allgroups<-Reduce(rbind,list.of.current.intake.impact.sims.unconsumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
CF.intake.impact.sims.unconsumed.allgroups<-Reduce(rbind,list.of.CF.intake.impact.sims.unconsumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

impact.sims.inedible.allgroups<-Reduce(rbind,list.of.impact.sims.inedible) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
substitution.impact.sims.inedible.allgroups<-Reduce(rbind,list.of.substitution.impact.sims.inedible) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
combined.impact.sims.inedible.allgroups<-Reduce(rbind,list.of.combined.impact.sims.inedible) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
delta.sims.inedible.allgroups<-Reduce(rbind,list.of.delta.sims.inedible) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

```

```
impact.sims.wasted.allgroups<-Reduce(rbind,list.of.impact.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
substitution.impact.sims.wasted.allgroups<-Reduce(rbind,list.of.substitution.impact.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
combined.impact.sims.wasted.allgroups<-Reduce(rbind,list.of.combined.impact.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
delta.sims.wasted.allgroups<-Reduce(rbind,list.of.delta.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

current.intake.impact.sims.inedible.allgroups<-Reduce(rbind,list.of.current.intake.impact.sims.inedible
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
CF.intake.impact.sims.inedible.allgroups<-Reduce(rbind,list.of.CF.intake.impact.sims.inedible) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
current.intake.impact.sims.wasted.allgroups<-Reduce(rbind,list.of.current.intake.impact.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
CF.intake.impact.sims.wasted.allgroups<-Reduce(rbind,list.of.CF.intake.impact.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
```

Save dataframes into csv files.

```
#setwd("C:\\Users\\fcudhe01\\Box\\Lasting_aim_3\\data\\out\\example_cost\\per_capita")
setwd(output_location)
fwrite(x=impact.sims.percapita.total.allgroups, file="per_capita/costenv.sims.percapita.granular.csv")
fwrite(x=substitution.impact.sims.percapita.total.allgroups, file="per_capita/substitution.costenv.sims.percapita.granular.csv")
fwrite(x=combined.impact.percapita.total.allgroups, file="per_capita/combined.costenv.sims.percapita.granular.csv")
fwrite(x=delta.percapita.allgroups, file="per_capita/delta.costenv.sims.percapita.granular.csv")

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.consumed.granular.csv")
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.sims.percapita.consumed.granular.csv")
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita.consumed.granular.csv")

fwrite(x=impact.sims.percapita.unconsumed.allgroups, file="per_capita/costenv.sims.percapita.unconsumed.granular.csv")
fwrite(x=substitution.impact.sims.percapita.unconsumed.allgroups, file="per_capita/substitution.costenv.sims.percapita.unconsumed.granular.csv")
fwrite(x=combined.impact.percapita.unconsumed.allgroups, file="per_capita/combined.costenv.sims.percapita.unconsumed.granular.csv")

fwrite(x=current.intake.impact.sims.percapita.total.allgroups, file="per_capita/current.intake.costenv.sims.percapita.total.allgroups.csv")
fwrite(x=CF.intake.impact.sims.percapita.total.allgroups, file="per_capita/CF.intake.costenv.sims.percapita.total.allgroups.csv")
fwrite(x=current.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/current.intake.costenv.sims.percapita.consumed.allgroups.csv")
fwrite(x=CF.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/CF.intake.costenv.sims.percapita.consumed.allgroups.csv")
fwrite(x=current.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/current.intake.costenv.sims.percapita.unconsumed.allgroups.csv")
fwrite(x=CF.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/CF.intake.costenv.sims.percapita.unconsumed.allgroups.csv")

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.inedible.granular.csv")
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.sims.percapita.inedible.granular.csv")
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita.inedible.granular.csv")

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.wasted.granular.csv")
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.sims.percapita.wasted.granular.csv")
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita.wasted.granular.csv")

fwrite(x=current.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/current.intake.costenv.sims.percapita.consumed.allgroups.csv")
fwrite(x=CF.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/CF.intake.costenv.sims.percapita.consumed.allgroups.csv")
fwrite(x=current.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/current.intake.costenv.sims.percapita.unconsumed.allgroups.csv")
fwrite(x=CF.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/CF.intake.costenv.sims.percapita.unconsumed.allgroups.csv")
```

Get per capita sims for all outputs of interest.

```
### per capita
```

```
impact.sims.percapita.total.allgroups<-get.percapita.sims(impact.sims=impact.sims.total.allgroups,
                                                         population.sims=population.sims.allgroups,
                                                         pop.sims.names=paste("X", c(1:n.sims), sep=""),
                                                         new.pop.sims.names=paste("pop", c(1:n.sims), sep="")
                                                         sims.names=paste("X", c(1:n.sims), sep=""))

substitution.impact.sims.percapita.total.allgroups<-get.percapita.sims(impact.sims=substitution.impact.
                                                         population.sims=population.sims.allgro
                                                         pop.sims.names=paste("X", c(1:n.sims),
                                                         new.pop.sims.names=paste("pop", c(1:n.
                                                         sims.names=paste("X", c(1:n.sims), sep

combined.impact.percapita.total.allgroups<-get.percapita.sims(impact.sims=combined.impact.sims.total.al
                                                         population.sims=population.sims.allgroups,
                                                         pop.sims.names=paste("X", c(1:n.sims), sep=""),
                                                         new.pop.sims.names=paste("pop", c(1:n.sims), se
                                                         sims.names=paste("X", c(1:n.sims), sep=""))

delta.percapita.allgroups<-get.percapita.sims(impact.sims=delta.sims.total.allgroups,
                                                         population.sims=population.sims.allgroups,
                                                         pop.sims.names=paste("X", c(1:n.sims), sep=""),
                                                         new.pop.sims.names=paste("pop", c(1:n.sims), sep=""),
                                                         sims.names=paste("X", c(1:n.sims), sep=""))

impact.sims.percapita.consumed.allgroups<-get.percapita.sims(impact.sims=impact.sims.consumed.allgroups
                                                         population.sims=population.sims.allgroups,
                                                         pop.sims.names=paste("X", c(1:n.sims), sep=""),
                                                         new.pop.sims.names=paste("pop", c(1:n.sims), sep="")
                                                         sims.names=paste("X", c(1:n.sims), sep=""))

substitution.impact.sims.percapita.consumed.allgroups<-get.percapita.sims(impact.sims=substitution.impa
                                                         population.sims=population.sims.allgro
                                                         pop.sims.names=paste("X", c(1:n.sims),
                                                         new.pop.sims.names=paste("pop", c(1:n.
                                                         sims.names=paste("X", c(1:n.sims), sep

combined.impact.percapita.consumed.allgroups<-get.percapita.sims(impact.sims=combined.impact.sims.consu
                                                         population.sims=population.sims.allgroups,
                                                         pop.sims.names=paste("X", c(1:n.sims), sep=""),
                                                         new.pop.sims.names=paste("pop", c(1:n.sims), se
                                                         sims.names=paste("X", c(1:n.sims), sep=""))

impact.sims.percapita.unconsumed.allgroups<-get.percapita.sims(impact.sims=impact.sims.unconsumed.allgr
                                                         population.sims=population.sims.allgroups,
                                                         pop.sims.names=paste("X", c(1:n.sims), sep
                                                         new.pop.sims.names=paste("pop", c(1:n.sims
                                                         sims.names=paste("X", c(1:n.sims), sep="")

substitution.impact.sims.percapita.unconsumed.allgroups<-get.percapita.sims(impact.sims=substitution.im
                                                         population.sims=population.si
                                                         pop.sims.names=paste("X", c(1
                                                         new.pop.sims.names=paste("pop
                                                         sims.names=paste("X", c(1:n.s

combined.impact.percapita.unconsumed.allgroups<-get.percapita.sims(impact.sims=combined.impact.sims.unc
                                                         population.sims=population.sims.allgro
                                                         pop.sims.names=paste("X", c(1:n.sims),
                                                         new.pop.sims.names=paste("pop", c(1:n.
                                                         sims.names=paste("X", c(1:n.sims), sep
```

```

current.intake.impact.sims.percapita.total.allgroups<-get.percapita.sims(impact.sims=current.intake.impact.sims,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))
CF.intake.impact.sims.percapita.total.allgroups<-get.percapita.sims(impact.sims=CF.intake.impact.sims,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))
current.intake.impact.sims.percapita.consumed.allgroups<-get.percapita.sims(impact.sims=current.intake.impact.sims,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))
CF.intake.impact.sims.percapita.consumed.allgroups<-get.percapita.sims(impact.sims=CF.intake.impact.sims,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))
current.intake.impact.sims.percapita.unconsumed.allgroups<-get.percapita.sims(impact.sims=current.intake.impact.sims,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))
CF.intake.impact.sims.percapita.unconsumed.allgroups<-get.percapita.sims(impact.sims=CF.intake.impact.sims,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))

impact.sims.percapita.inedible.allgroups<-get.percapita.sims(impact.sims=impact.sims.inedible.allgroups,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))
substitution.impact.sims.percapita.inedible.allgroups<-get.percapita.sims(impact.sims=substitution.impact.sims,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))
combined.impact.percapita.inedible.allgroups<-get.percapita.sims(impact.sims=combined.impact.sims.inedible.allgroups,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))

impact.sims.percapita.wasted.allgroups<-get.percapita.sims(impact.sims=impact.sims.wasted.allgroups,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))
substitution.impact.sims.percapita.wasted.allgroups<-get.percapita.sims(impact.sims=substitution.impact.sims,
    population.sims=population.sims.allgroups,
    pop.sims.names=paste("X", c(1:n.sims), sep="")
    new.pop.sims.names=paste("pop", c(1:n.sims),
    sims.names=paste("X", c(1:n.sims), sep=""))

```

```

population.sims=population.si
pop.sims.names=paste("X", c(1
new.pop.sims.names=paste("pop
sims.names=paste("X", c(1:n.s
combined.impact.percapita.wasted.allgroups<-get.percapita.sims(impact.sims=combined.impact.sims.wasted.
population.sims=population.sims.allgro
pop.sims.names=paste("X", c(1:n.sims),
new.pop.sims.names=paste("pop", c(1:n.
sims.names=paste("X", c(1:n.sims), sep

current.intake.impact.sims.percapita.inedible.allgroups<-get.percapita.sims(impact.sims=current.intake.
population.sims=population.
pop.sims.names=paste("X", c
new.pop.sims.names=paste("p
sims.names=paste("X", c(1:n

CF.intake.impact.sims.percapita.inedible.allgroups<-get.percapita.sims(impact.sims=CF.intake.impact.sim
population.sims=population.sims.
pop.sims.names=paste("X", c(1:n.
new.pop.sims.names=paste("pop",
sims.names=paste("X", c(1:n.sims

current.intake.impact.sims.percapita.wasted.allgroups<-get.percapita.sims(impact.sims=current.intake.im
population.sims=populatio
pop.sims.names=paste("X",
new.pop.sims.names=paste(
sims.names=paste("X", c(1

CF.intake.impact.sims.percapita.wasted.allgroups<-get.percapita.sims(impact.sims=CF.intake.impact.sims.
population.sims=population.sim
pop.sims.names=paste("X", c(1:
new.pop.sims.names=paste("pop"
sims.names=paste("X", c(1:n.si

```

Save them into csv files.



```
#setwd("C:\\Users\\fcudhe01\\Box\\Lasting_aim_3\\data\\out\\example_cost\\per_capita")
setwd(output_location)
fwrite(x=impact.sims.percapita.total.allgroups, file="per_capita/costenv.sims.percapita.granular.csv")
fwrite(x=substitution.impact.sims.percapita.total.allgroups, file="per_capita/substitution.costenv.sims.percapita.granular.csv")
fwrite(x=combined.impact.percapita.total.allgroups, file="per_capita/combined.costenv.sims.percapita.granular.csv")
fwrite(x=delta.percapita.allgroups, file="per_capita/delta.costenv.sims.percapita.granular.csv")

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.consumed.granular.csv")
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.sims.percapita.consumed.granular.csv")
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita.consumed.granular.csv")

fwrite(x=impact.sims.percapita.unconsumed.allgroups, file="per_capita/costenv.sims.percapita.unconsumed.granular.csv")
fwrite(x=substitution.impact.sims.percapita.unconsumed.allgroups, file="per_capita/substitution.costenv.sims.percapita.unconsumed.granular.csv")
fwrite(x=combined.impact.percapita.unconsumed.allgroups, file="per_capita/combined.costenv.sims.percapita.unconsumed.granular.csv")

fwrite(x=current.intake.impact.sims.percapita.total.allgroups, file="per_capita/current.intake.costenv.sims.percapita.total.allgroups.csv")
fwrite(x=CF.intake.impact.sims.percapita.total.allgroups, file="per_capita/CF.intake.costenv.sims.percapita.total.allgroups.csv")
fwrite(x=current.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/current.intake.costenv.sims.percapita.consumed.allgroups.csv")
fwrite(x=CF.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/CF.intake.costenv.sims.percapita.consumed.allgroups.csv")
fwrite(x=current.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/current.intake.costenv.sims.percapita.unconsumed.allgroups.csv")
fwrite(x=CF.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/CF.intake.costenv.sims.percapita.unconsumed.allgroups.csv")

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.inedible.granular.csv")
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.sims.percapita.inedible.granular.csv")
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita.inedible.granular.csv")

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.wasted.granular.csv")
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.sims.percapita.wasted.granular.csv")
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita.wasted.granular.csv")

fwrite(x=current.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/current.intake.costenv.sims.percapita.consumed.allgroups.csv")
fwrite(x=CF.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/CF.intake.costenv.sims.percapita.consumed.allgroups.csv")
fwrite(x=current.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/current.intake.costenv.sims.percapita.unconsumed.allgroups.csv")
fwrite(x=CF.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/CF.intake.costenv.sims.percapita.unconsumed.allgroups.csv")
```

Get **summary stats** from simulations for all outcome of interests and save (for both total and per capita impact).

```
#####
##get summary stats
#####
all.impact.sims.summary<-get.summary.stats(impact.sims=impact.sims.total.allgroups,
                                           substitution.impact.sims=substitution.impact.sims.total.allg
                                           combined.impact.sims=combined.impact.sims.total.allgroups,
                                           delta.sims=delta.sims.total.allgroups,
                                           impact.sims.consumed=impact.sims.consumed.allgroups,
                                           substitution.impact.sims.consumed=substitution.impact.sims.c
                                           combined.impact.sims.consumed=combined.impact.sims.consumed.
                                           impact.sims.unconsumed=impact.sims.unconsumed.allgroups,
                                           substitution.impact.sims.unconsumed=substitution.impact.sims
                                           combined.impact.sims.unconsumed=combined.impact.sims.unconsu
                                           current.intake.impact.sims=current.intake.impact.sims.total.
                                           current.intake.impact.sims.consumed=current.intake.impact.si
                                           current.intake.impact.sims.unconsumed=current.intake.impact.
                                           CF.intake.impact.sims=CF.intake.impact.sims.total.allgroups,
                                           CF.intake.impact.sims.consumed=CF.intake.impact.sims.consume
                                           CF.intake.impact.sims.unconsumed=CF.intake.impact.sims.uncon
                                           impact.sims.inedible=impact.sims.inedible.allgroups,
                                           substitution.impact.sims.inedible=substitution.impact.sims.i
                                           combined.impact.sims.inedible=combined.impact.sims.inedible.
                                           impact.sims.wasted=impact.sims.wasted.allgroups,
                                           substitution.impact.sims.wasted=substitution.impact.sims.was
                                           combined.impact.sims.wasted=combined.impact.sims.wasted.allg
                                           current.intake.impact.sims.inedible=current.intake.impact.si
                                           current.intake.impact.sims.wasted=current.intake.impact.sims
                                           CF.intake.impact.sims.inedible=CF.intake.impact.sims.inedibl
                                           CF.intake.impact.sims.wasted=CF.intake.impact.sims.wasted.al
                                           vars=paste("X", 1:n.sims, sep=""))
#setwd("C:\\Users\\fcudhe01\\Box\\Lasting_aim_3\\data\\out\\example_cost")
setwd(output_location)
write.csv(x=all.impact.sims.summary, file="costenv.summary.granular.csv")

##
#now use the function for per capita results
all.impact.sims.percapita.summary<-get.summary.stats(impact.sims=impact.sims.percapita.total.allgroups,
                                                       substitution.impact.sims=substitution.impact.sims.
                                                       combined.impact.sims=combined.impact.percapita.tot
                                                       delta.sims=delta.percapita.allgroups,
                                                       impact.sims.consumed=impact.sims.percapita.consume
                                                       substitution.impact.sims.consumed=substitution.imp
                                                       combined.impact.sims.consumed=combined.impact.perc
                                                       impact.sims.unconsumed=impact.sims.percapita.uncon
                                                       substitution.impact.sims.unconsumed=substitution.i
                                                       combined.impact.sims.unconsumed=combined.impact.pe
                                                       current.intake.impact.sims=current.intake.impact.s
                                                       current.intake.impact.sims.consumed=current.intake
                                                       current.intake.impact.sims.unconsumed=current.inta
                                                       CF.intake.impact.sims=CF.intake.impact.sims.total.
                                                       CF.intake.impact.sims.consumed=CF.intake.impact.si
                                                       CF.intake.impact.sims.unconsumed=CF.intake.impact.
                                                       impact.sims.inedible=impact.sims.percapita.inedibl
                                                       substitution.impact.sims.inedible=substitution.imp
```

```

combined.impact.sims.inedible=combined.impact.perc
impact.sims.wasted=impact.sims.percapita.wasted.al
substitution.impact.sims.wasted=substitution.impact.percapita.wasted
combined.impact.sims.wasted=combined.impact.percapita.wasted
current.intake.impact.sims.inedible=current.intake.percapita.inedible
current.intake.impact.sims.wasted=current.intake.percapita.wasted
CF.intake.impact.sims.inedible=CF.intake.impact.percapita.inedible
CF.intake.impact.sims.wasted=CF.intake.impact.percapita.wasted
vars=paste("X", 1:n.sims, sep="")
#setwd("C:\\Users\\fcudhe01\\Box\\Lasting_aim_3\\data\\out\\example_cost\\per_capita")
setwd(output_location)
write.csv(x=all.impact.sims.percapita.summary, file="per_capita/costenv.percapita.summary.granular.csv")

```

Next, we want to get summary stat for all possible strata combinations (not just the most granular, age/sex/race/foodgroup).

First we create a list of all possible **strata** combinations. Note, the string vector “strata” contains all the strata being used for the analysis. If you want to use different strata, you need to change the line defining “strata” accordingly.

```

#####get sums based on less granular subgroups#####
##get all possible combinations of strata
strata<-c("age_gp", "sex_gp", "race_gp", "Foodgroup")

strata.combos<-list()
for(i in 1:length(strata))
{
  x<-combn(strata, i)
  combos.for.i<-split(x, rep(1:ncol(x), each = nrow(x))) #https://stackoverflow.com/questions/6819804/h
  strata.combos<-c(strata.combos, combos.for.i)
}
strata.combos["null"]<-list(NULL) ##add null element to list to use for overall numbers

```

Get summary stats by all strata combinations for all outcomes of interest. Note that the objects being created here are lists of lists. The second level are the different strata combinations. The first level denotes outcome type: 1 = total summary stats, 2 = total sims, 3 = per capita summary stats, 4 = per capita sims.

```

# summary.stats.by.strata<-function(impact.sims.allgroups, strata.combos.list, pop.sims)
# {
#   cols.to.sum<-paste("X", 1:n.sims, sep="")
#   pop.cols.to.sum<-paste("pop", 1:n.sims, sep="")
#
#   sims.data.table<-as.data.table(impact.sims.allgroups)
#   pop.sims.data.table<-as.data.table(pop.sims)
#
#   strata.combos.sims<-list()
#   strata.combos.sims.long<-list()
#   strata.combos.summary.stats.long<-list()
#   strata.combos.summary.stats.wide<-list()
#
#   pop.strata.combos.sims<-list()
#
#   percapita.sims<-list()
#   percapita.sims.long<-list()
#   percapita.summary.stats.long<-list()
#   percapita.summary.stats.wide<-list()
#
#   for(j in 1:length(strata.combos))
#   {
#     labels<-c(strata.combos[[j]], "intake_units", "outcome", "outcome_unit")
#
#     strata.combos.sims[[j]]<-sims.data.table[,lapply(.SD, sum), by=c(strata.combos[[j]], "intake_unit",
#                               .SDcols=cols.to.sum)]
#
#     pop.strata<-strata.combos[[j]][!(strata.combos[[j]] %in% c("Foodgroup", "broad_foodgroup"))]
#
#     if(is.null(pop.strata))
#     {
#       pop.strata.combos.sims[[j]]<-pop.sims.data.table[,lapply(.SD, sum), .SDcols=cols.to.sum]
#     }else
#     {
#       pop.strata.combos.sims[[j]]<-pop.sims.data.table[,lapply(.SD, sum), by=c(pop.strata), .SDcols=c(
#         names(pop.strata.combos.sims[[j]])<-gsub(x=names(pop.strata.combos.sims[[j]]), pattern="X", repla
#
#       if(length(intersect(names(strata.combos.sims[[j]]), names(pop.strata.combos.sims[[j]])))==0)
#       {
#         merged<-cbind(strata.combos.sims[[j]], pop.strata.combos.sims[[j]])
#       }
#       if(length(intersect(names(strata.combos.sims[[j]]), names(pop.strata.combos.sims[[j]])))!=0)
#       {
#         merged<-merge(strata.combos.sims[[j]], pop.strata.combos.sims[[j]])
#       }
#
#       merged.names<-merged[,..labels]
#       percapita.sims[[j]]<-merged[,..cols.to.sum]/merged[,..pop.cols.to.sum]
#       percapita.sims[[j]]<-cbind(merged.names, percapita.sims[[j]])
#
#       strata.combos.sims.long[[j]]<-melt(strata.combos.sims[[j]], id.vars = c(strata.combos[[j]], "inta
#       strata.combos.summary.stats.long[[j]]<-strata.combos.sims.long[[j]][, .(quantile(value, c(.025, .

```

```

#   strata.combos.summary.stats.long[[j]]<-cbind(strata.combos.summary.stats.long[[j]],
#   as.factor(rep(c("Lower_bound (2.5th percentile)", "m
#   #strata.combos.summary.stats.wide[[j]]<-dcast(strata.combos.summary.stats.long[[j]], "Comorbid_co
#   strata.combos.summary.stats.wide[[j]]<-dcast(strata.combos.summary.stats.long[[j]], paste(paste(s
#   value.var="V1")
#
#   ##nperhaps in future, make it so that the function doesn't rely on specific names for intake_unit
#
#   percapita.sims.long[[j]]<-melt(percipita.sims[[j]], id.vars = c(strata.combos[[j]], "intake_units
#   percapita.summary.stats.long[[j]]<-percipita.sims.long[[j]][, .(quantile(value, c(.025, .5, .975)
#   percapita.summary.stats.long[[j]]<-cbind(percipita.summary.stats.long[[j]],
#   as.factor(rep(c("Lower_bound (2.5th percentile)", "media
#   dim(percipita.summary.stats.long[[j]])[1]/
#   #strata.combos.summary.stats.wide[[j]]<-dcast(strata.combos.summary.stats.long[[j]], "Comorbid_co
#   percapita.summary.stats.wide[[j]]<-dcast(percipita.summary.stats.long[[j]], paste(paste(strata.co
#   value.var="V1")
#
#
#   }
#
#   return(list("summary.output" = strata.combos.summary.stats.wide,
#   "full.sims.output" = strata.combos.sims,
#   "summary.output.percapita" = percapita.summary.stats.wide,
#   "full.sims.output.percapita" = percapita.sims)
#   )
# }

setwd(paste(output_location, "/By_SubGroup", sep=""))
summ.stats.by.strata.cost<-summary.stats.by.strata(impact.sims.allgroups=impact.sims.total.allgroups,
strata.combos.list=strata.combos,
pop.sims=pop.sims)
summ.stats.by.strata.substitution.cost<-summary.stats.by.strata(impact.sims.allgroups=substitution.impa
strata.combos.list=strata.combos,
pop.sims=pop.sims)
summ.stats.by.strata.combined.cost<-summary.stats.by.strata(impact.sims.allgroups=combined.impact.sims.
strata.combos.list=strata.combos,
pop.sims=pop.sims)
summ.stats.by.strata.delta_forcost<-summary.stats.by.strata(impact.sims.allgroups=delta.sims.total.allg
strata.combos.list=strata.combos,
pop.sims=pop.sims)

summ.stats.by.strata.combined.consumed.cost<-summary.stats.by.strata(impact.sims.allgroups=combined.imp
strata.combos.list=strata.combos,
pop.sims=pop.sims)
summ.stats.by.strata.combined.unconsumed.cost<-summary.stats.by.strata(impact.sims.allgroups=combined.i
strata.combos.list=strata.combos,
pop.sims=pop.sims)

summ.stats.by.strata.current.intake.cost<-summary.stats.by.strata(impact.sims.allgroups=current.intake.
strata.combos.list=strata.combos,
pop.sims=pop.sims)
summ.stats.by.strata.CF.intake.cost<-summary.stats.by.strata(impact.sims.allgroups=CF.intake.impact.sim
strata.combos.list=strata.combos
pop.sims=pop.sims)
summ.stats.by.strata.current.intake.consumed.cost<-summary.stats.by.strata(impact.sims.allgroups=curren

```

```

strata.combos.list=strata.combos,
pop.sims=pop.sims)
summ.stats.by.strata.CF.intake.consumed.cost<-summary.stats.by.strata(impact.sims.allgroups=CF.intake.i
strata.combos.list=strata.combos,
pop.sims=pop.sims)
summ.stats.by.strata.current.intake.unconsumed.cost<-summary.stats.by.strata(impact.sims.allgroups=curr
strata.combos.list=strata.co
pop.sims=pop.sims)
summ.stats.by.strata.CF.intake.unconsumed.cost<-summary.stats.by.strata(impact.sims.allgroups=CF.intake
strata.combos.list=strata.combos,
pop.sims=pop.sims)

##inedible and wasted
summ.stats.by.strata.combined.inedible.cost<-summary.stats.by.strata(impact.sims.allgroups=combined.imp
strata.combos.list=strata.combos,
pop.sims=pop.sims)
summ.stats.by.strata.combined.wasted.cost<-summary.stats.by.strata(impact.sims.allgroups=combined.impac
strata.combos.list=strata.combos
pop.sims=pop.sims)

summ.stats.by.strata.current.intake.inedible.cost<-summary.stats.by.strata(impact.sims.allgroups=curren
strata.combos.list=strata.co
pop.sims=pop.sims)
summ.stats.by.strata.CF.intake.inedible.cost<-summary.stats.by.strata(impact.sims.allgroups=CF.intake.i
strata.combos.list=strata.combos,
pop.sims=pop.sims)
summ.stats.by.strata.current.intake.wasted.cost<-summary.stats.by.strata(impact.sims.allgroups=current.
strata.combos.list=strata.
pop.sims=pop.sims)
summ.stats.by.strata.CF.intake.wasted.cost<-summary.stats.by.strata(impact.sims.allgroups=CF.intake.imp
strata.combos.list=strata.combo
pop.sims=pop.sims)

```

Next, we're going to go through a for loop and output all outcomes into their own csv files for each strata combination.

First, start the for loop

```

for(j in 1:length(strata.combos))
{

```

First, save full sim outputs for total and per capita

```
setwd(paste(output_location, "/By_SubGroup/full_sims", sep=""))
#setwd("C:\\Users\\fcudhe01\\Box\\Lasting_aim_3\\data\\out\\example_cost\\BySubGroup\\full_sims")
fwrite(x=summ.stats.by.strata.cost[[2]][[j]], file=paste("full.costenv.sims.output_by_", paste(strata
fwrite(x=summ.stats.by.strata.substitution.cost[[2]][[j]], file=paste("full.costenv.sims.substitution
fwrite(x=summ.stats.by.strata.combined.cost[[2]][[j]], file=paste("full.costenv.sims.combined.output_

setwd(paste(output_location, "/per_capita/By_SubGroup/full_sims", sep=""))
#setwd("C:\\Users\\fcudhe01\\Box\\Lasting_aim_3\\data\\out\\example_cost\\per_capita\\BySubGroup\\ful
fwrite(x=summ.stats.by.strata.cost[[4]][[j]], file=paste("full.costenv.sims.percapita.output_by_", pa
fwrite(x=summ.stats.by.strata.substitution.cost[[4]][[j]], file=paste("full.costenv.sims.percapita.su
fwrite(x=summ.stats.by.strata.combined.cost[[4]][[j]], file=paste("full.costenv.sims.percapita.combin
```

**Next**, create outcome names to be used as variable names, and put relevant outcome data frames into two lists (for total and per capita).

*#need to edit all this renaming code so it renames the new mean and SD columns. Perhaps take the oppu  
## having a set vector to rename, and using cat to combine the var name and the old name to create new*

```
to.rename<-c("lower_bound (2.5th percentile)", "median", "upper_bound (97.5th percentile)", "mean",
outcome.names<-c("impact",
                  "substitution_impact",
                  "combined_impact",
                  "delta",
                  "combined_consumed_impact",
                  "combined_unconsumed_impact",
                  "current_intake_impact",
                  "CF_intake_impact",
                  "current_intake_consumed_impact",
                  "CF_intake_consumed_impact",
                  "current_intake_unconsumed_impact",
                  "CF_intake_unconsumed_impact",
                  ##inedible, wasted
                  "combined_inedible_impact",
                  "combined_wasted_impact",
                  "current_intake_inedible_impact",
                  "CF_intake_inedible_impact",
                  "current_intake_wasted_impact",
                  "CF_intake_wasted_impact"
                )
```

```
outcome.data<-list(summ.stats.by.strata.cost[[1]][[j]],
                  summ.stats.by.strata.substitution.cost[[1]][[j]],
                  summ.stats.by.strata.combined.cost[[1]][[j]],
                  summ.stats.by.strata.delta_forcast[[1]][[j]],
                  summ.stats.by.strata.combined_consumed.cost[[1]][[j]],
                  summ.stats.by.strata.combined_unconsumed.cost[[1]][[j]],
                  summ.stats.by.strata.current_intake.cost[[1]][[j]],
                  summ.stats.by.strata.CF_intake.cost[[1]][[j]],
                  summ.stats.by.strata.current_intake_consumed.cost[[1]][[j]],
                  summ.stats.by.strata.CF_intake_consumed.cost[[1]][[j]],
                  summ.stats.by.strata.current_intake_unconsumed.cost[[1]][[j]],
                  summ.stats.by.strata.CF_intake_unconsumed.cost[[1]][[j]],
                  ##inedible, wasted
                  summ.stats.by.strata.combined_inedible.cost[[1]][[j]],
                  summ.stats.by.strata.combined_wasted.cost[[1]][[j]],
                  summ.stats.by.strata.current_intake_inedible.cost[[1]][[j]],
                  summ.stats.by.strata.CF_intake_inedible.cost[[1]][[j]],
                  summ.stats.by.strata.current_intake_wasted.cost[[1]][[j]],
                  summ.stats.by.strata.CF_intake_wasted.cost[[1]][[j]]
                )
```

```
outcome.data.per.capita<-list(summ.stats.by.strata.cost[[3]][[j]],
                              summ.stats.by.strata.substitution.cost[[3]][[j]],
                              summ.stats.by.strata.combined.cost[[3]][[j]],
                              summ.stats.by.strata.delta_forcast[[3]][[j]],
                              summ.stats.by.strata.combined_consumed.cost[[3]][[j]],
                              summ.stats.by.strata.combined_unconsumed.cost[[3]][[j]],
                              summ.stats.by.strata.current_intake.cost[[3]][[j]],
                              summ.stats.by.strata.CF_intake.cost[[3]][[j]],
                              summ.stats.by.strata.current_intake_consumed.cost[[3]][[j]],
                              summ.stats.by.strata.CF_intake_consumed.cost[[3]][[j]],
```



```

summ.stats.by.strata.current.intake.unconsumed.cost[[3]][[j]],
summ.stats.by.strata.CF.intake.unconsumed.cost[[3]][[j]],
##inedible, wasted
summ.stats.by.strata.combined.inedible.cost[[3]][[j]],
summ.stats.by.strata.combined.wasted.cost[[3]][[j]],
summ.stats.by.strata.current.intake.inedible.cost[[3]][[j]],
summ.stats.by.strata.CF.intake.inedible.cost[[3]][[j]],
summ.stats.by.strata.current.intake.wasted.cost[[3]][[j]],
summ.stats.by.strata.CF.intake.wasted.cost[[3]][[j]]
)

for(k in 1:length(outcome.names))
{
  names(outcome.data[[k]])[which(names(outcome.data[[k]]) %in% c("lower_bound (2.5th percentile)", "me
    paste(outcome.names[k], c("lower_bound (2.5th percentile)", "median", "upper_bound (97.5th percen
names(outcome.data.per.capita[[k]])[which(names(outcome.data.per.capita[[k]]) %in% c("lower_bound (2
    paste(outcome.names[k], c("lower_bound (2.5th percentile)", "median", "upper_bound (97.5th percen
}

# names(summ.stats.by.strata.cost[[1]][[j]])[which(names(summ.stats.by.strata.cost[[1]][[j]]) %in% c(
#   c("impact_lower_bound (2.5th percentile)", "impact_median", "impact_upper_bound (97.5th percenti
# names(summ.stats.by.strata.substitution.cost[[1]][[j]])[which(names(summ.stats.by.strata.substituti
#   c("substitution_impact_lower_bound (2.5th percentile)", "substitution_impact_median", "substituti
# names(summ.stats.by.strata.combined.cost[[1]][[j]])[which(names(summ.stats.by.strata.combined.cost[
#   c("combined_impact_lower_bound (2.5th percentile)", "combined_impact_median", "combined_impact_up
# names(summ.stats.by.strata.delta_forcast[[1]][[j]])[which(names(summ.stats.by.strata.delta_forcast[
#   c("delta_lower_bound (2.5th percentile)", "delta_median", "delta_upper_bound (97.5th percentile)
# names(summ.stats.by.strata.combined.consumed.cost[[1]][[j]])[which(names(summ.stats.by.strata.combi
#   c("combined_consumed_impact_lower_bound (2.5th percentile)", "combined_consumed_impact_median", "
# names(summ.stats.by.strata.combined.unconsumed.cost[[1]][[j]])[which(names(summ.stats.by.strata.com
#   c("combined_unconsumed_impact_lower_bound (2.5th percentile)", "combined_unconsumed_impact_median
# names(summ.stats.by.strata.current.intake.cost[[1]][[j]])[which(names(summ.stats.by.strata.current.
#   c("current_intake_impact_lower_bound (2.5th percentile)", "current_intake_impact_median", "curren
# names(summ.stats.by.strata.CF.intake.cost[[1]][[j]])[which(names(summ.stats.by.strata.CF.intake.cos
#   c("CF_intake_impact_lower_bound (2.5th percentile)", "CF_intake_impact_median", "CF_intake_impact
# names(summ.stats.by.strata.current.intake.consumed.cost[[1]][[j]])[which(names(summ.stats.by.strata
#   c("current_intake_consumed_impact_lower_bound (2.5th percentile)", "current_intake_consumed_impac
# names(summ.stats.by.strata.CF.intake.consumed.cost[[1]][[j]])[which(names(summ.stats.by.strata.CF.i
#   c("CF_intake_consumed_impact_lower_bound (2.5th percentile)", "CF_intake_consumed_impact_median",
# names(summ.stats.by.strata.current.intake.unconsumed.cost[[1]][[j]])[which(names(summ.stats.by.stra
#   c("current_intake_unconsumed_impact_lower_bound (2.5th percentile)", "current_intake_unconsumed_i
# names(summ.stats.by.strata.CF.intake.unconsumed.cost[[1]][[j]])[which(names(summ.stats.by.strata.CF
#   c("CF_intake_unconsumed_impact_lower_bound (2.5th percentile)", "CF_intake_unconsumed_impact_medi
#
#
#####
#
# names(summ.stats.by.strata.cost[[3]][[j]])[which(names(summ.stats.by.strata.cost[[3]][[j]]) %in% c(
#   c("impact_lower_bound (2.5th percentile)", "impact_median", "impact_upper_bound (97.5th percenti
# names(summ.stats.by.strata.substitution.cost[[3]][[j]])[which(names(summ.stats.by.strata.substituti
#   c("substitution_impact_lower_bound (2.5th percentile)", "substitution_impact_median", "substituti
# names(summ.stats.by.strata.combined.cost[[3]][[j]])[which(names(summ.stats.by.strata.combined.cost[
#   c("combinedimpact_lower_bound (2.5th percentile)", "combined_impact_median", "combined_impact_upp
# names(summ.stats.by.strata.delta_forcast[[3]][[j]])[which(names(summ.stats.by.strata.delta_forcast[
#   c("delta_lower_bound (2.5th percentile)", "delta_median", "delta_upper_bound (97.5th percentile)

```

```
#
# names(summ.stats.by.strata.combined.consumed.cost[[3]][[j]])[which(names(summ.stats.by.strata.combined.consumed.cost[[3]][[j]]) %in% c("combined_consumed_impact_lower_bound (2.5th percentile)", "combined_consumed_impact_median", "combined_consumed_impact_upper_bound (2.5th percentile)"))
# names(summ.stats.by.strata.combined.unconsumed.cost[[3]][[j]])[which(names(summ.stats.by.strata.combined.unconsumed.cost[[3]][[j]]) %in% c("combined_unconsumed_impact_lower_bound (2.5th percentile)", "combined_unconsumed_impact_median", "combined_unconsumed_impact_upper_bound (2.5th percentile)"))
# names(summ.stats.by.strata.current.intake.cost[[3]][[j]])[which(names(summ.stats.by.strata.current.intake.cost[[3]][[j]]) %in% c("current_intake_impact_lower_bound (2.5th percentile)", "current_intake_impact_median", "current_intake_impact_upper_bound (2.5th percentile)"))
# names(summ.stats.by.strata.CF.intake.cost[[3]][[j]])[which(names(summ.stats.by.strata.CF.intake.cost[[3]][[j]]) %in% c("CF_intake_impact_lower_bound (2.5th percentile)", "CF_intake_impact_median", "CF_intake_impact_upper_bound (2.5th percentile)"))
# names(summ.stats.by.strata.current.intake.consumed.cost[[3]][[j]])[which(names(summ.stats.by.strata.current.intake.consumed.cost[[3]][[j]]) %in% c("current_intake_consumed_impact_lower_bound (2.5th percentile)", "current_intake_consumed_impact_median", "current_intake_consumed_impact_upper_bound (2.5th percentile)"))
# names(summ.stats.by.strata.CF.intake.consumed.cost[[3]][[j]])[which(names(summ.stats.by.strata.CF.intake.consumed.cost[[3]][[j]]) %in% c("CF_intake_consumed_impact_lower_bound (2.5th percentile)", "CF_intake_consumed_impact_median", "CF_intake_consumed_impact_upper_bound (2.5th percentile)"))
# names(summ.stats.by.strata.current.intake.unconsumed.cost[[3]][[j]])[which(names(summ.stats.by.strata.current.intake.unconsumed.cost[[3]][[j]]) %in% c("current_intake_unconsumed_impact_lower_bound (2.5th percentile)", "current_intake_unconsumed_impact_median", "current_intake_unconsumed_impact_upper_bound (2.5th percentile)"))
# names(summ.stats.by.strata.CF.intake.unconsumed.cost[[3]][[j]])[which(names(summ.stats.by.strata.CF.intake.unconsumed.cost[[3]][[j]]) %in% c("CF_intake_unconsumed_impact_lower_bound (2.5th percentile)", "CF_intake_unconsumed_impact_median", "CF_intake_unconsumed_impact_upper_bound (2.5th percentile)"))
# ````
```

Then,for both lists, merge their contents into a data frame.

```
list.names.intersect<-Reduce(intersect, lapply(outcome.data, colnames))

all.summ.stats.by.strata<-Reduce(function(...) merge(..., by=list.names.intersect), outcome.data)
all.summ.stats.by.strata.per.capita<-Reduce(function(...) merge(..., by=list.names.intersect), outcome.data)
# 
# all.summ.stats.by.strata<-Reduce(function(...) merge(..., all = T),list(summ.stats.by.strata.delta[,c("delta","cost","substition","combination")],
#                                     summ.stats.by.strata.cost[,c("cost","substition","combination")],
#                                     summ.stats.by.strata.substitution[,c("substition","combination")],
#                                     summ.stats.by.strata.combination[,c("combination")],
#                                     summ.stats.by.strata.current[,c("current","CF.intensity")],
#                                     summ.stats.by.strata.CF.intensity[,c("CF.intensity","current")],
#                                     summ.stats.by.strata.current[,c("current","CF.intensity")],
#                                     summ.stats.by.strata.CF.intensity[,c("CF.intensity")])
#                                     )
#                                     )
# all.summ.stats.by.strata.per.capita<-Reduce(function(...) merge(..., all = T),list(summ.stats.by.strata.per.capita[,c("per.capita","delta","cost","substition","combination")],
#                                             summ.stats.by.strata.per.capita[,c("per.capita","cost","substition","combination")],
#                                             summ.stats.by.strata.per.capita[,c("per.capita","substition","combination")],
#                                             summ.stats.by.strata.per.capita[,c("per.capita","combination")],
#                                             summ.stats.by.strata.per.capita[,c("per.capita","current","CF.intensity")],
#                                             summ.stats.by.strata.per.capita[,c("per.capita","CF.intensity","current")],
#                                             summ.stats.by.strata.per.capita[,c("per.capita","current","CF.intensity")],
#                                             summ.stats.by.strata.per.capita[,c("per.capita","CF.intensity")])
#                                             )
#                                             )
```

Save, the two outputs created (one for total and one for per capita) as csvs. Each of these dataframes will have summary stats for all outputs of interest for subgroup j.

```
setwd(paste(output_location, "/By_SubGroup", sep=""))
#setwd("C:\\Users\\fcudhe01\\Box\\Lasting_aim_3\\data\\out\\example_cost\\BySubGroup")
write.csv(x=all.summ.stats.by.strata, file=paste("summary.output_by_", paste(strata.combos[[j]], coll

setwd(paste(output_location, "/per_capita/By_SubGroup", sep=""))
#setwd("C:\\Users\\fcudhe01\\Box\\Lasting_aim_3\\data\\out\\example_cost\\per_capita\\BySubGroup")
write.csv(x=all.summ.stats.by.strata.per.capita, file=paste("summary.output.percapita_by_", paste(str
```

End loop.

```
}
```